

 Reliable INSTITUTE Unleashing Potential	MATHEMATICS REVISION PLAN-3 Target : JEE (Advanced)-2023	DPP DAILY PRACTICE PROBLEM DPP NO. # 02 (Advanced)
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Max. Time : 129 Minutes	Total Marks-166
Q. 01 to 06 Comprehension ('-1' negative marking) [3 Marks,3 Min.][18,18]	
Q. 07 to 26 Multiple choice objective ('-1' negative marking) [4 Marks,3 Min.][80,60]	
Q. 27 to 28 Match the column (no negative marking) [4 Marks,3 min.][8,6]	
Q. 29 to 43 Numerical Type ('-1' negative marking) [4 Marks,3 Min.][60,45]	

DPP Syllabus : Indefinite Integration, Definite Integration & its Application and Differential Equations

Comprehension # 1 (Q.No. 1 to 2)

For a real valued function $f(x, k) = \lim_{n \rightarrow \infty} \left(\frac{k}{(x)^{1/n} + k - 1} \right)^n \forall x \in (0, 1), k > 1, n \in \mathbb{N}$ and let

$$T(k) = \int_0^1 f(x, k) \cdot (\ln x)^2 dx, \text{ then}$$

- The value of $\int_0^1 f\left(x, \frac{5}{2}\right) dx$ is
 (A) 5 (B) $\frac{5}{2}$ (C) $\frac{5}{3}$ (D) $\frac{2}{5}$
- The value of $T(3)$ is
 (A) 3 (B) $\frac{9}{4}$ (C) $\frac{27}{4}$ (D) 27

Comprehension # 2 (Q.No. 3 to 4)

Let $f : A \rightarrow B$ $f(x) = \frac{x+a}{bx^2+cx+2}$, where A represents domain set and B represents range set of function

$f(x)$, $a, b, c \in \mathbb{R}$, $f(-1) = 0$ and $y = 1$ is an asymptote of $y = f(x)$ and $y = g(x)$ is the inverse of $f(x)$.

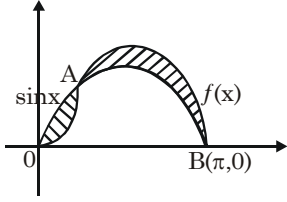
- Area bounded between the curves $y = f(x)$ and $y = g(x)$ is
 (A) $2\sqrt{5} + \ln \left(\frac{3-\sqrt{5}}{3+\sqrt{5}} \right)$ (B) $3\sqrt{5} + 2 \ln \left(\frac{3+\sqrt{5}}{3-\sqrt{5}} \right)$
 (C) $3\sqrt{5} + 4 \ln \left(\frac{3-\sqrt{5}}{3+\sqrt{5}} \right)$ (D) $3\sqrt{5} + 2 \ln \left(\frac{3-\sqrt{5}}{3+\sqrt{5}} \right)$
- Area of region enclosed by asymptotes of curves $y = f(x)$ and $y = g(x)$ is :
 (A) 4 (B) 9 (C) 12 (D) 25

Comprehension # 3 (Q.No. 5 to 6)

A tank contains 100 litre of salt solution which has 100 grams of salt at time $t = 0$. It has an inlet valve through which salt solution of 10 gm/litre concentration enters the tank at the rate of 10 ltr/hr. Tank also has an outflow valve through which salt solution in tank exists at a rate of 5 ltr/hr. The contents of the tank are kept thoroughly mixed at all times.

The tank can only hold upto 200 litre of contents, the rest exist out through a spill over valve

5. Amount of salt (in grams) in tank after 20 hours is
 (A) 2000 (B) 1850 (C) 1750 (D) 1550
6. Amount of salt (in grams) in tank at $t = 20 \ln(10e)$ will be
 (A) 1955 (B) 1895 (C) 1925 (D) none of these
7. Let $\int \frac{e^x(x-1)(x-\ln x)}{x^2} dx = e^x \left(1 - \frac{1+f(x)}{x} \right) + C$, then
 (A) $f(1) = 0$ (B) $f(xy) = f(x) + f(y) \forall x \in \mathbb{R}^+$
 (C) $f(x) = x$ has no solution (D) $f'(1) = 1$
8. If $\int \frac{8x^{43} + 13x^{38}}{(x^{13} + x^5 + 1)^4} dx = \frac{ax^m}{(x^{13} + x^5 + 1)^n} + k$ (where 'k' is the constant of integration)
 (A) $m + n = 43$ (B) $m + n = 42$ (C) $\frac{m+n}{a} = 126$ (D) $\frac{m+n}{a} = 129$
9. If $f(x) = \int (e^{18x} + e^{10x} + e^{6x}) \sqrt{e^{12x} + 2e^{4x} + 4} dx$ and $f(0) = \frac{7\sqrt{7}}{24}$, then
 (A) $\lim_{x \rightarrow \infty} \frac{x^{16}}{f(x)} = 12$ (B) $\lim_{x \rightarrow \infty} \frac{x^{16}}{f(x)} = 0$ (C) $\lim_{x \rightarrow \infty} \frac{f(x)}{e^{24x}} = \frac{1}{24}$ (D) $\lim_{x \rightarrow \infty} \frac{f(x)}{e^{24x}} = \frac{1}{12}$
10. Let $I_n = \int \frac{x^n dx}{\sqrt{x^2 + 2x + 5}}$, then $I_n = \left(\frac{x^{m-1}}{n} \right) \sqrt{x^2 + 2x + 5} - \lambda I_{n-1} - \mu I_{n-2}$;
 where λ and μ are functions of 'n', then
 (A) $m = n - 1$ (B) $\lambda(2) = \frac{3}{2}$ (C) $\lambda(1) = 1$ (D) $\mu(5) = 4$
11. If $\int_0^{\pi/2} \frac{1 - \sin 2x}{(1 + \sin 2x)^2} dx = \frac{a}{b}$ (where a, b are relatively prime), then.
 (A) $a + b + ab = 7$ (B) $a + b = 4$ (C) $a - b = 2$ (D) $a + b + ab = 4$

12. If $f(x) = -\int_0^x \log(\cos t) dt$ and $f(x) - 2f\left(\frac{\pi}{4} + \frac{x}{2}\right) + 2f\left(\frac{\pi}{4} - \frac{x}{2}\right) = g(x)$, then
- (A) $g(x)$ is a quadratic function
 (B) Area bounded by $y = |g(x)|$ and $y = 1$ is $\log_2 e$
 (C) Number of solution of equation $g(x) = \operatorname{sgn}(x)$ is 1
 (D) Number of solution of equation $g(x) = \{x\}$ is 3 (where $\{.\}$ represents fractional part function)
13. Let $J = \int_0^1 \frac{x}{1+x^8} dx$. Then which of the following is/are correct :
- (A) $J > \frac{1}{4}$ (B) $J < \frac{\pi}{8}$ (C) $J < \frac{1}{4}$ (D) $J > \frac{\pi}{8}$
14. Let $y = f(x)$ and $y = \sin x$ intersect at origin, $A(a, \sin a)$, $B(\pi, 0)$ as shown in figure and $A_1 = 1 + (a - 1) \cos a - \sin a$ is area bounded by curves $y = f(x)$ and $y = \sin x$ between $x = 0$ and $x = a$, $A_2 =$ Area bounded by curves $y = f(x)$, $y = \sin x$ between $x = a$ and $x = \pi$, then
- 
- (A) The function $f(x)$ is $x \sin x$ (B) $A_2 = (\pi - \sin 1 - 1)$ sq. unit
 (C) Value of $(A_2 + A_1)$ is $(\pi - 2 \sin 1)$ sq. unit (D) Value of $(A_2 - A_1)$ is $(\pi - 2)$ sq. unit
15. If $I_n = \int_0^\pi \frac{\sin n\theta}{\sin \theta} d\theta$, $n \in \mathbb{N}$, then
- (A) $I_n - I_{n-2} = 0$ ($n \geq 4$) (B) $I_n - I_{n-2} = \pi$ ($n \geq 4$) (C) $|I_{50} - I_{29}| = \pi$ (D) $I_{50} - I_{29} = 0$
16. Let $f(x) = \int_1^x \frac{3^t}{1+t^2} dt$, where $x > 0$, then which is true ?
- (A) for $0 < \alpha < \beta$, $f(\alpha) < f(\beta)$ (B) for $0 < \alpha < \beta$, $f'(\alpha) < f'(\beta)$
 (C) $f(x) + \frac{\pi}{4} \leq \tan^{-1} x \forall x \geq 1$ (D) $f(x) + \frac{\pi}{4} \geq \tan^{-1} x \forall x \geq 1$
17. Let A_1 and A_2 are two real numbers such that $A_1 = \int_0^{\pi/4} \sin(\tan x) dx$ and $A_2 = \int_0^{\pi/3} \tan(\sin x) dx$, then for A_1 and A_2 which of the following statements is(are) TRUE?
- (A) $A_1 < \frac{\sqrt{2}-1}{\sqrt{2}}$ (B) $A_2 \geq \ln \frac{4}{3}$ (C) $A_1 \geq \frac{\pi^2}{16}$ (D) $A_2 \leq \ln 2$

18. Let $y = g(x)$ be a real valued function and $g'(x)$ is continuous function such that $g'(x) \geq 0$ and $g(0) = 0$, $g(1) = 1$ and if area bounded by $y = \sqrt{1 + (g'(x))^2}$, $x = 0$, $x = 1$ and x -axis is S , then which of the following statements is(are) TRUE?
- (A) $S < 2$ (B) $S > 2$ (C) $S < \frac{1}{\sqrt{2}}$ (D) $S \geq \sqrt{2}$
19. $T_n = \sum_{r=2n}^{3n-1} \frac{r}{r^2 + n^2}$, $S_n = \sum_{r=2n+1}^{3n} \frac{r}{r^2 + n^2}$, then $\forall n \in \{1, 2, 3, \dots\}$:
- (A) $T_n > \frac{1}{2} \ln 2$ (B) $S_n < \frac{1}{2} \ln 2$ (C) $T_n < \frac{1}{2} \ln 2$ (D) $S_n > \frac{1}{2} \ln 2$
20. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(x) = x + \int_0^x f(t) dt$, for all $x \in \mathbb{R}$. Then
- (A) The number of solutions of equation $f(x) = 0$ is/are equal to 1
 (B) The number of solutions of equation $f(x) = x$ is/are equal to 1
 (C) The number of solutions of equation $f(x) = f^{-1}(x)$ is/are equal to 0
 (D) $\lim_{x \rightarrow -\infty} f(x) = -1$
21. Let $y'(x) \cdot g(x) - y(x) \cdot g'(x) + y^2(x) = 0$, $y(-1) = 1$, $x \in \mathbb{R}$ where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(-1) = 0$ then
- (A) $\frac{g(2)}{y(2)} = 3$ (B) $\int_1^2 \frac{g(x)}{y(x)} dx = \frac{5}{2}$
- (C) $\int_1^2 \left(\frac{g(x)}{y(x)} \cdot \frac{1}{x^2 \sqrt{x^2 + \left(\frac{g(x)}{y(x)} \right)^2}} \right) dx = 2\sqrt{5} - \sqrt{13}$ (D) $\int_1^2 \left(\frac{g(x)}{y(x)} \cdot \frac{1}{x^2 \sqrt{x^2 + \left(\frac{g(x)}{y(x)} \right)^2}} \right) dx = \frac{7}{2(2\sqrt{5} + 13)}$
22. Equation of curve passing through (3, 4) and satisfying the differential equation, $y \left(\frac{dy}{dx} \right)^2 + (x - y) \frac{dy}{dx} - x = 0$ can be
- (A) $x^2 + y^2 = 25$ (B) $x^2 - y^2 + 7 = 0$ (C) $xy = 12$ (D) $x - y + 1 = 0$
23. If curve represented by differential equation $\frac{dy}{dx} = y^2 \tan^2 x + y^2 - 2y \tan x$ passes through (0,1) then
- (A) $\frac{dy}{dx}$ can be explicitly expressed as $\frac{2(x-1)\sin 2x + 1 + \cos 2x}{2(1-x)^2}$
- (B) $\frac{dy}{dx}$ can be explicitly expressed as $\frac{2(x-1)\sin 2x - \cos 2x}{2(1-x)^2}$
- (C) the curve passes through the point $\left(\frac{\pi}{2}, 0 \right)$
- (D) the curve passes through the point $(2\pi, 0)$

24. Let $y = f(x)$ be a real valued differentiable function $\forall x \in \mathbb{R}$, such that $\frac{f'(x)}{f(x)} \neq -1$ and

$(f'(x))^3 + x^3 + 3xf(x)f'(x) = (f(x))^3 \forall x \in \mathbb{R}$, then which of the following statements is(are) TRUE?

- (A) If $f(-1) = 0$, then $f(x) = (x+1)e^k$ (B) If $f(-1) = 1$, then $f(x) = (x+1) - (e^{x-1})$
 (C) $\frac{f'(c) - f(c)}{c} = -1$ for some $c \in \mathbb{R}$ (D) $\frac{f'(c) - f(c)}{c} = e$ for some $c \in \mathbb{R}$

25. Solution of the differential equation $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$ is

- (A) $x^2 - y^2 + cx(x-y) = 0$ (B) $x^2 + y^2 + c(x+y) = 0$
 (C) a straight line if it passes through $(1, -1)$ (D) a circle if it passes through $(1, 1)$

26. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \leq x \leq 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$.

Which of the following hold(s) good?

- (A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$

27. Match the following $\int f(x)$ is equal to, if

Column-I

- (A) $f(x) = \frac{1}{(x^2+1)\sqrt{x^2+2}}$
 (B) $f(x) = \frac{1}{(x+2)\sqrt{x^2+6x+7}}$
 (C) $f(x) = \frac{x^4+x^8}{(1-x^4)^{7/2}}$
 (D) $f(x) = \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}$

Column-II

- (p) $\frac{x^5}{5(1-x^4)^{5/2}} + C$
 (q) $\sin^{-1}\left(\frac{x+1}{(x+2)\sqrt{2}}\right) + C$
 (r) $(\sqrt{x}-2)\sqrt{1-x} + \cos^{-1}\sqrt{x} + C$
 (s) $-\tan^{-1}\sqrt{1+\frac{2}{x^2}} + C$
 (t) $\frac{x^6}{6(1-x^4)^{5/2}} + C$

28. **Column-I**

- (A) Solution of $\left(\frac{dy}{dx}\right)\left(\frac{d^3y}{dx^3}\right) - 3\left(\frac{d^2y}{dx^2}\right)^2 = 0$ is
 (B) Solution of $(x^2y^2 - 1)dy + 2xy^3dx = 0$ is
 (C) Solution of $\frac{dy}{dx} = xy(x^2y^2 - 1)$ is
 (D) Solution of $\frac{dy}{dx}(x^2y^3 + xy) = 1$ is

Column-II

- (p) $y^2(x^2 + 1 + ce^{x^2}) = 1$
 (q) $x = A_1y^2 + A_2y + A_3$
 (r) $\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$
 (s) $x^2y^2 + 1 = cy$

29. Let $I = \int \frac{3e^x + 5\sin x + 10\cos x}{e^x + 4\sin x + 3\cos x} dx = ax + b \ln(e^x + c \sin x + d \cos x) + C$, where C is constant of integration, then the value of $(a^2 + b^2 + c^2 + d^2)$ is equal to
30. Let $f(x) = \int x^2 \cos^2 x (2x + 6 \tan x - 2x \tan^2 x) dx$ and $f(x)$ passes through the point $(\pi, 0)$, then the number of solution of the equation $f(x) = x^3$ in $[0, 2\pi]$ is :
31. If $f(x) = \int \frac{x^2 - 1}{x\sqrt{x^4 + 4x^3 + 22x^2 + 4x + 1}} dx$ and $f(1) = \ln(4 + 4\sqrt{2})$, then $[f(3)]$ is equal to : (where $[.]$ represents greatest integer function)
32. The number of integer value/s of 'x' satisfying $\int_0^{2[x+14]} \left\{ \frac{x}{2} \right\} dx = \int_0^{\{x\}} [x + 14] dx$ is/are :
(where $[.]$ and $\{.\}$ denote greatest integer and fractional part functions respectively.)
33. If $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 2x \cos^2 x \\ x^2 & \sec x & \sin x + x^3 \\ 1 & 2 & x + \tan x \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} (x^2 + 1)(f(x) + f''(x)) dx$ is
34. If $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx = \frac{5\pi^2}{\lambda}$; then ' λ ' is
35. Let 'f' be a differentiable function such that $f(f(x)) = x$, $\forall x \in [0, 1]$. Suppose $f(0) = 1$; then :
 $6063 \int_0^1 (x - f(x))^{2020} dx =$
36. Find the value of $\frac{29 \int_0^1 (1 - x^4)^7 dx}{4 \int_0^1 (1 - x^4)^6 dx}$.
37. If $\lim_{n \rightarrow \infty} \left(\log \left(\prod_{r=1}^n \left(\frac{n+ar}{n} \right) \right) \cdot \sum_{r=0}^{n-1} \left(\frac{e^{r/2n}}{n} \right)^2 \right) = 1$, then $[a]$ is equal to (where $[.]$ denotes greatest integer function)
38. If $\lim_{n \rightarrow \infty} n^{-\frac{1}{2} \left(1 + \frac{1}{n} \right)} (1^1 \cdot 2^2 \cdot 3^3 \cdots n^n)^{\frac{1}{n^2}} = P$, then $\log P$ is equal to
39. Let $y(x)$ be solution of differential equation $\frac{dy}{dx} = \frac{x^2 y^3 + 2y^3 - 2xy^2}{y^2 x^3 + 2x^2 y - 2x^3}$. if $y(1) = 1$ and the solution is $y = x e^{k_1 \left(\frac{x-y}{xy} \right)^{k_2}}$, then $k_1 + k_2$ equals

40. A tangent is drawn at the point (x_i, y_i) to curve $y = f(x)$, which intersects the x-axis at $(x_{i+1}, 0)$. Now again a tangent is drawn at (x_{i+1}, y_{i+1}) to the curve which intersects the x-axis at $(x_{i+2}, 0)$ and the process is repeated 'n' times. If $x_1, x_2, x_3, \dots$ form an AP with common difference $\log_2 e$ and the curve passing through $(0, 2)$ & $(-2, k)$, then k is equal to
41. Find the sum of order and degree of differential equation of all the conics touching the y-axis at origin and having centre on the x-axis
42. Let $y(x)$ be the solution of differential equation $(e^{x^2} + e^{y^2})y \frac{dy}{dx} + e^{x^2}(xy^2 - x) = 0$, if $y(0) = 0$, then $\left| y \left(\sqrt{\ln \frac{4}{3} + \frac{1}{4}} \right) \right|$ is equal to
43. A curve $y = f(x)$ passes through the origin and lies entirely in the first quadrant. Through any point $P(x, y)$ on the curve, lines are drawn parallel to the coordinate axes. If the curve divides the area formed by these lines and coordinate axes in 3 : 2 (where $3k$ is the area of upper part and $2k$ is the area of lower part), then the value of $\frac{f(4) + f(16)}{f(25)}$ is equal to

ANSWER KEY

1. (C) 2. (C) 3. (D) 4. (B) 5. (D) 6. (A) 7. (ABCD)
8. (BC) 9. (BC) 10. (BCD) 11. (AB) 12. (BCD) 13. (AD) 14. (ABCD)
15. (AC) 16. (ABD) 17. (BCD) 18. (AD) 19. (AB) 20. (ABD) 21. (ABD)
22. (AD) 23. (AC) 24. (BC) 25. (BCD) 26. (ABD)
27. $(A) \rightarrow (s), (B) \rightarrow (q), (C) \rightarrow (p), (D) \rightarrow (r)$
28. $(A) \rightarrow (q), (B) \rightarrow (s), (C) \rightarrow (p), (D) \rightarrow (r)$
29. 30 30. 3 31. 2 32. 1 33. 0 34. 160 35. 3
36. 07.00 37. 1 38. (-0.25) 39. 1 40. 8 41. (3) 42. (0.5)
43. (0.576)